

Convex Analysis and Monotone Operator Theory

Chapter 27: Fermat's Rule in Convex Optimization

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2017

Chapter Overview

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Minimizers and Subdifferentials I

Fermat's Rule (Fundamental Theorem):

Let H be a real Hilbert space and $f \in \Gamma_0(H)$. Then:

$$x \in \operatorname{Argmin}(f) \iff 0 \in \partial f(x)$$

This characterizes minimizers as zeros of the subdifferential.

Key Insight:

- For convex differentiable functions: $0 \in \partial f(x) \iff \nabla f(x) = 0$
- For non-smooth functions: subdifferential contains all subgradients
- Works in infinite-dimensional Hilbert spaces
- No regularity assumptions needed

Examples:

- 1 Quadratic function: $f(x) = \frac{1}{2}\|x\|^2$ has $\operatorname{Argmin}(f) = \{0\}$
- 2 Indicator function: $f(x) = \iota_C(x)$ (0 if $x \in C$, ∞ otherwise)
- 3 Norm: $f(x) = \|x\|$ has $\operatorname{Argmin}(f) = \{0\}$

Theorem 27.1: General Characterizations I

Theorem 27.1: Let $f, g \in \Gamma_0(H)$ and $L \in \mathcal{B}(H, K)$ where K is a real Hilbert space. The following statements are equivalent:

- (i) $\bar{x}$ is a solution to $\min_{x \in H} f(x) + g(Lx)$
- (ii) $\bar{x} \in \text{zer}(\partial f + L^* \circ (\partial g) \circ L)$
- (iii) $(\exists v \in \partial g(L\bar{x})) -L^*v \in \partial f(\bar{x})$
- (iv) $(\exists v \in \partial g(L\bar{x}))(\forall y \in H) \langle \bar{x} - y \mid L^*v \rangle + f(\bar{x}) \leq f(y)$

Moreover, when g is Gâteaux differentiable at $L\bar{x}$:

- (v) $-L^*(\nabla g(L\bar{x})) \in \partial f(\bar{x})$
- (vi) $(\forall y \in H) \langle \bar{x} - y \mid L^*(\nabla g(L\bar{x})) \rangle + f(\bar{x}) \leq f(y)$
- (vii) $(\forall \gamma \in \mathbb{R}_{++}) \bar{x} = \text{Prox}_{\gamma f}(\bar{x} - \gamma L^*(\nabla g(L\bar{x})))$
- (viii) $f(\bar{x}) + f^*(-L^*(\nabla g(L\bar{x}))) + \langle L\bar{x} \mid \nabla g(L\bar{x}) \rangle = 0$

Applications of Fermat's Rule I

Unconstrained Minimization:

If f is convex and differentiable, then:

$$\text{Argmin}(f) = \{x : \nabla f(x) = 0\}$$

Constrained Minimization over Convex Sets:

Let C be a closed convex set and f be differentiable. Then:

$$\bar{x} \in \text{Argmin}_{x \in C} f(x) \iff \bar{x} \in C \text{ and } (\forall y \in C) \langle y - \bar{x} \mid \nabla f(\bar{x}) \rangle \geq 0$$

This is the **variational inequality** characterization.

Composite Optimization:

For $\min_x f(x) + g(x)$ where f, g are convex:

$$0 \in \partial f(\bar{x}) + \partial g(\bar{x})$$

Proposition 27.8: Constrained Minimization I

Proposition 27.8: Let C be a nonempty closed convex subset of H , $f \in \Gamma_0(H)$, $\bar{x} \in H$, and $\gamma \in \mathbb{R}_{++}$. Suppose one of the following regularity conditions holds:

- (a) $0 \in \text{sri}(C - \text{dom } f)$
- (b) H is finite-dimensional, C is polyhedral, and $C \cap \text{ri}(\text{dom } f) \neq \emptyset$
- (c) H is finite-dimensional, C is a polyhedral set, f is polyhedral, and $C \cap \text{dom } f \neq \emptyset$

Then these are equivalent:

- (i) $\bar{x}$ is a solution to $\min_{x \in C} f(x)$
- (ii) $\bar{x} \in \text{zer}(N_C + \partial f)$
- (iii) $\bar{x} \in \text{Prox}_f(\text{Fix}(2P_C - \text{Id}) \circ (2\text{Prox}_f - \text{Id}))$
- (iv) $(\exists u \in N_C \bar{x}) -u \in \partial f(\bar{x})$
- (v) $(\exists u \in \partial f(\bar{x})) -u \in N_C \bar{x}$
- (vi) $\bar{x} \in C$ and $(\exists u \in \partial f(\bar{x}))(\forall y \in C) \langle \bar{x} - y \mid u \rangle \leq 0$

When f is Gâteaux differentiable at $\bar{x}$, additional equivalences:

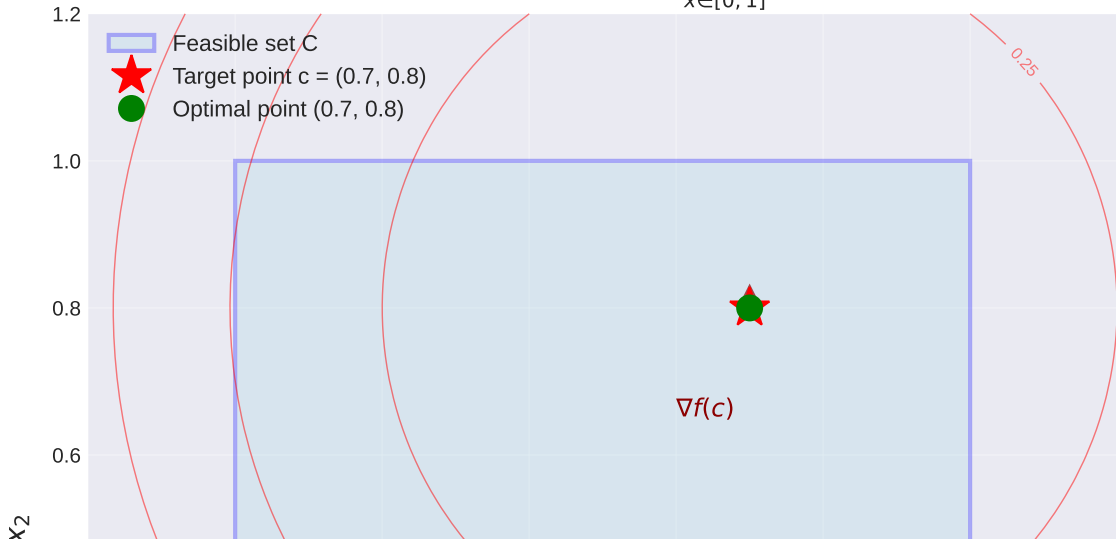
- $-\nabla f(\bar{x}) \in N_C \bar{x}$
- $\bar{x} \in C$ and $(\forall y \in C) \langle \bar{x} - y \mid \nabla f(\bar{x}) \rangle \leq 0$
- $\bar{x} = P_C(\bar{x} - \gamma \nabla f(\bar{x}))$

Constrained Optimization: Python Example

```
1 import numpy as np
2 from scipy.optimize import minimize
3 import matplotlib.pyplot as plt
4
5 # Minimize  $\|x - c\|^2$  subject to  $x$  in  $[0, 1]^2$ 
6 c = np.array([0.7, 0.8])
7 x0 = np.array([0.0, 0.0])
8
9 # Objective:  $\|x - c\|^2$ 
10 def f(x):
11     return np.sum((x - c)**2)
12
13 def grad_f(x):
14     return 2 * (x - c)
15
16 # Constraint:  $x$  in  $[0, 1]^2$ 
17 bounds = [(0, 1), (0, 1)]
18
19 # Solve
20 result = minimize(f, x0, method='L-BFGS-B',
21                  jac=grad_f, bounds=bounds)
22 x_opt = result.x
23
24 print(f"Optimal solution: {x_opt}")
25 print(f"Gradient at optimum: {grad_f(x_opt)}")
26
27 # Check:  $x_{\text{opt}}$  = projection of  $c$  onto  $[0, 1]^2$ 
```

Visualization: Constrained Optimization

Constrained Minimization: $\min_{x \in [0, 1]^2} \|x - c\|^2$



Affine Constraints: Lagrange Multipliers I

Problem with Affine Constraints:

Consider the problem:

$$\min_{x \in H} f(x) \quad \text{subject to} \quad Lx = r$$

where $f \in \Gamma_0(H)$, $L \in \mathcal{B}(H, K)$, and $r \in L(\text{dom } f)$.

Proposition 27.14: $\bar{x}$ is a solution if and only if:

$$L\bar{x} = r \quad \text{and} \quad (\exists \bar{v} \in K) \quad -L^*\bar{v} \in \partial f(\bar{x})$$

In this case, $\bar{v}$ is a **Lagrange multiplier**.

When f is Gâteaux differentiable:

$$L\bar{x} = r \quad \text{and} \quad (\exists \bar{v} \in K) \quad \nabla f(\bar{x}) = -L^*\bar{v}$$

Key Insight: The solution satisfies both the constraint and a subdifferential condition involving Lagrange multipliers.

Affine Constraints: Python Example

```
1 import numpy as np
2 from scipy.optimize import minimize
3
4 # Minimize  $\|x\|^2$  subject to  $x_1 + x_2 + x_3 = 1$ 
5 # Exact solution:  $x = (1/3, 1/3, 1/3)$ 
6
7 def f(x):
8     return np.sum(x**2)
9
10 def grad_f(x):
11     return 2 * x
12
13 # Affine constraint:  $Ax = b$ 
14 A = np.array([[1, 1, 1]])
15 b = np.array([1.0])
16
17 def constraint(x):
18     return A @ x - b
19
20 x0 = np.array([0.0, 0.0, 0.0])
21
22 # Solve with constraint
23 from scipy.optimize import LinearConstraint
24 lc = LinearConstraint(A, b, b)
25 result = minimize(f, x0, jac=grad_f, constraints=lc)
26
27 x_opt = result.x
```

Special Cases of Affine Constraints I

Case 1: Finite-Dimensional Non-negativity Constraints

Let $A \in \mathbb{R}^{M \times N}$, $f \in \Gamma_0(\mathbb{R}^N)$. Consider:

$$\min_{x \in \mathbb{R}_+^N} f(x) \quad \text{subject to} \quad Ax \in \mathbb{R}_+^M$$

The KKT conditions characterize solutions.

Case 2: Linear Programming Constraints

For polyhedral sets C and D with $C \cap L^{-1}(D) \neq \emptyset$:

$$\min_{x \in C} f(x) + g(Lx)$$

where $f = \iota_C$ and $g = \iota_D$.

The solution satisfies normal cone conditions.

Corollary 27.15 (Closed Convex Sets):

Special Cases of Affine Constraints II

For closed convex sets C and D with $f = \iota_C$, $g = \iota_D$:

$$x \in \mathbb{R}^M \text{ and } (\exists u \in K^\perp \cap \partial f(\bar{x})) \langle \bar{x} - y \mid u \rangle \leq 0 \quad (\forall y \in C)$$

Composite Functions: Corollary 27.3 I

Corollary 27.3: Let $f, g \in \Gamma_0(H)$, $\bar{x} \in H$, and $\gamma \in \mathbb{R}_{++}$.

Suppose one of these conditions holds:

- (a) $0 \in \text{sri}(\text{dom } g - \text{dom } f)$
- (b) H finite-dimensional, g polyhedral, $\text{dom } g \cap \text{ri}(\text{dom } f) \neq \emptyset$
- (c) H finite-dimensional, f, g polyhedral, $\text{dom } f \cap \text{dom } g \neq \emptyset$

Then these are equivalent:

- (i) $\bar{x}$ is a solution to $\min_x f(x) + g(x)$
- (ii) $\bar{x} \in \text{zer}(\partial f + \partial g)$
- (iii) $\bar{x} \in \text{Prox}_g(\text{Fix}(2\text{Prox}_f - \text{Id}) \circ (2\text{Prox}_g - \text{Id}))$
- (iv) $(\exists u \in \partial g(\bar{x})) -u \in \partial f(\bar{x})$
- (v) $(\exists u \in \partial f(\bar{x})) -u \in \partial g(\bar{x})$
- (vi) $(\exists u \in \partial g(\bar{x}))(\forall y \in H) \langle \bar{x} - y \mid u \rangle + f(\bar{x}) \leq f(y)$

When g is Gâteaux differentiable at $\bar{x}$:

- $-\nabla g(\bar{x}) \in \partial f(\bar{x})$
- $(\forall y \in H) \langle \bar{x} - y \mid \nabla g(\bar{x}) \rangle + f(\bar{x}) \leq f(y)$
- $\bar{x} = \text{Prox}_{\gamma f}(\bar{x} - \gamma \nabla g(\bar{x}))$

Composite Optimization: Python Example

```
1 import numpy as np
2 from scipy.optimize import minimize
3 from scipy.sparse.linalg import cg
4
5 # Minimize  $\|x - x_0\|^2 + \lambda \|x\|_1$ 
6 # (quadratic + sparsity-inducing term)
7 # Solution: soft thresholding
8
9 x0 = np.array([1.5, 0.3, -2.0, 0.1, 0.8])
10 lam = 0.5
11
12 def f(x):
13     return np.sum((x - x0)**2)
14
15 def f_sparse(x):
16     return np.sum((x - x0)**2) + lam * np.sum(np.abs(x))
17
18 # Solve
19 result = minimize(f_sparse, x0, method='L-BFGS-B')
20 x_opt = result.x
21
22 # Soft thresholding:  $x^* = \text{sign}(x_0) * \max(|x_0| - \lambda/2, 0)$ 
23 x_analytical = np.sign(x0) * np.maximum(np.abs(x0) - lam/2, 0)
24
25 print(f"Numerical solution: {x_opt}")
26 print(f"Analytical (soft-thresh): {x_analytical}")
27 print(f"Match: {np.allclose(x_opt, x_analytical, atol=1e-4)}")
```

Example 27.10: Projection onto a Convex Set I

Problem: Project a point p onto a closed convex set C .

$$\min_{x \in C} \|x - p\|^2$$

Solution: The projection $\bar{x} = P_C(p)$ is characterized by:

$$\langle p - \bar{x} \mid y - \bar{x} \rangle \leq 0 \quad \forall y \in C$$

This means $(p - \bar{x})$ is orthogonal to C at $\bar{x}$.

Verification with Fermat's Rule:

- Objective: $f(x) = \frac{1}{2}\|x - p\|^2$
- Gradient: $\nabla f(x) = x - p$
- Normal cone to C at $\bar{x}$: $N_C(\bar{x})$
- Condition: $-\nabla f(\bar{x}) = p - \bar{x} \in N_C(\bar{x})$ (correct)

Example 27.10: Python Visualization

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.spatial import ConvexHull
4
5 # Define a convex polygon
6 vertices = np.array([[0, 0], [2, 0], [2.5, 1.5],
7                      [1, 2], [0, 1.5]])
8 hull = ConvexHull(vertices)
9
10 # Point to project
11 p = np.array([1.5, -0.5])
12
13 # Simple projection by checking distance to vertices
14 # and edges (numerical approach)
15 from scipy.optimize import minimize
16
17 def dist_sq(x):
18     return np.sum((x - p)**2)
19
20 # Start from a point in the hull
21 x0 = np.mean(vertices, axis=0)
22
23 # Constraint: point in convex hull
24 # Use linear constraints from convex hull equations
25 A_ub, b_ub = [], []
26 for simplex in hull.simplices:
27     # For each edge/facet, create inequality constraint
```


Example 27.11 and 27.12: Bilinear Forms I

Example 27.11 (Stampacchia): Let $F : H \times H \rightarrow \mathbb{R}$ be bilinear with:

$$(\forall x, y \in H) \quad |F(x, y)| \leq \beta \|x\| \|y\| \quad \text{and} \quad F(x, x) \geq \alpha \|x\|^2$$

For a nonempty closed convex set C and $\ell \in \mathcal{B}(H, \mathbb{R})$:

$$\exists! \bar{x} \in C : (\forall y \in C) \quad F(\bar{x}, y - \bar{x}) \geq \ell(y - \bar{x})$$

Applications:

- Variational inequalities
- Nonlinear elasticity problems
- Convex optimization with indefinite quadratic objectives

Example 27.12 (Lax-Milgram): Similar structure but for optimization:

$$\min_{x \in C} \left\{ \frac{1}{2} F(x, x) - \ell(x) \right\}$$

The unique minimizer satisfies the variational inequality.

Key Theorems and Results I

Main Contributions:

- 1 **Fermat's Rule:** $x^* \in \operatorname{Argmin}(f) \iff 0 \in \partial f(x^*)$
- 2 **Composite Minimization:** Conditions for minimizing $f(x) + g(Lx)$
- 3 **Constrained Minimization:** Characterizations using normal cones
- 4 **Affine Constraints:** Lagrange multiplier conditions
- 5 **Variational Inequalities:** Connection to monotone operators
- 6 **Regularity Conditions:** Sufficient conditions for equivalences

Practical Implications:

- First-order optimality conditions for nonsmooth convex optimization
- Foundation for proximal algorithms
- Characterization of solutions without explicit computation
- Extension to infinite-dimensional settings

Connections to Other Chapters

- **Subdifferentials** (Ch. 16): Necessary tools for stating conditions
- **Monotone Operators** (Ch. 20): ∂f is maximally monotone
- **Resolvents** (Ch. 23): Connection to proximal point algorithm
- **Projection Operators** (Ch. 24): Special case of minimization
- **Splitting Methods** (Ch. 22): Algorithms based on Fermat's rule
- **Proximal Algorithms** (Ch. 28): Direct application to minimization